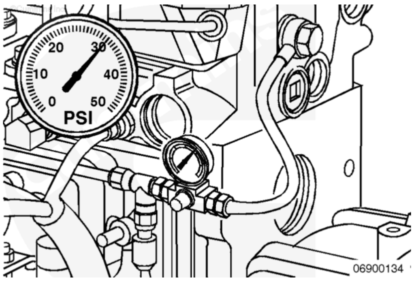
Trainings

Test

B4.5 RGT Engines

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : This step only applies to B4.5 RGT engines. B3.9, B4.5 and B5.9 engines do not have an internal fuel drain line in the cylinder head.

If troubleshooting coolant in the fuel, fuel in the coolant, fuel in the oil, or oil in the fuel, pressurize the internal fuel drain line in the cylinder head and check for leaks.

Remove the fuel drain line at the back of the cylinder head. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-021-tr.html)

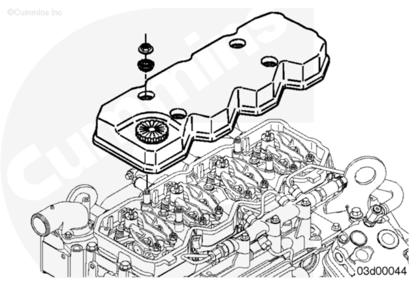
Connect a regulated air supply hose to the cylinder head fuel drain port with a shut off valve on the air supply side of the pressure gauge.

Apply air pressure.

Air Pressure		
kpa		psi
276	NOM	40

Shut off the air supply to the fuel drain port and monitor the pressure gauge reading. The pressure should hold steady. If the pressure drops rapidly, check for leaks around the:

- Test fittings
- Fuel connectors at the cylinder head.



Remove the rocker lever cover and check for air bubbles around the injectors. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)

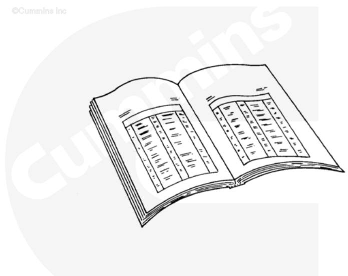
Remove the radiator cap and check for air bubbles in the cooling system.

If the source of the leak can **not** be determined, remove the cylinder head and pressure test the complete cylinder head. See the Pressure Test step of this procedure. Replace the cylinder head, if necessary.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.


ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

If removing the cylinder head with the injectors installed, be careful not to damage the tips of the injector. Do not set the cylinder head down on the combustion face with the injectors installed. Damage to the injector tips will result.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-009.html)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Remove any original equipment manufacturer (OEM) accessories attached to the cylinder head. See the equipment manufacturer service information.
- Disconnect the air crossover connection. Refer to Procedure 010-019 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- Disconnect all water and heater hoses attached to the cylinder head. See the equipment manufacturer service information.

Note : Omit the following steps if the engine is not equipped with the component or if it is not necessary to remove the component to remove the cylinder head. For some components, it is only necessary to remove the component if the cylinder head is being replaced or rebuilt.

- Remove the fuel filter. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html)
- Remove the fuel filter head. Refer to Procedure 006-017 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-017-tr.html)
- Remove the air intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-023-tr.html)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Remove the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Remove the fan hub assembly. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-036-tr.html)
- Loosen the alternator link, mounting bolt, and water inlet connection capscrews. Remove the alternator bracket mounting capscrews and pivot the alternator away from

the engine. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-001-tr.html)

- Remove the alternator bracket from the thermostat housing. Refer to Procedure 013-003 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-003-tr.html)
- Remove the thermostat housing and thermostat from the engine. Refer to Procedure 008-013 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-013-tr.html)

Note : The following components are required to be removed in order to remove the cylinder head.

- If equipped, remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/40/40-011-007-tr.html)
- Remove the rocker lever cover(s). Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the rocker lever housing and gasket(s). Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-013-tr.html)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Remove the crosshead, if equipped. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-001-tr.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Remove the fuel supply lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html)
- Remove the injector supply lines (high-pressure). Refer to Procedure 006-051 in Section 6.

(/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)

- If B4.5 RGT engines, remove the fuel connectors (head-mounted). Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-052-tr.html)

Note : For B4.5 RGT engines, do not remove the injectors at this time. Remove the cylinder head with the injectors installed so that injector protrusion can be checked.

- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)

Remove

B4.5 RGT Engines

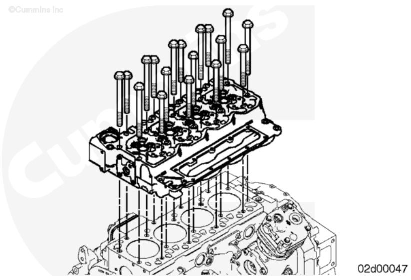
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

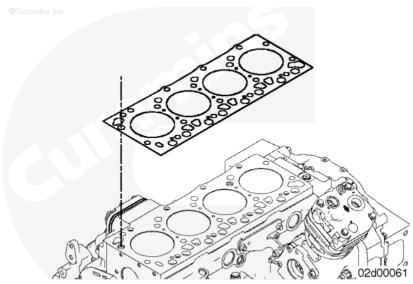
⚠ CAUTION ⚠

If removing the cylinder head with the injectors installed, be careful not to damage the tips of the injector. Do not set the cylinder head down on the combustion face with the injectors installed. Damage to the injector tips will result.

Remove the cylinder head capscrews and cylinder head.

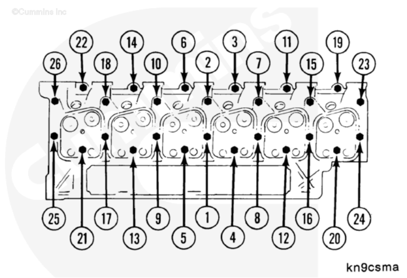


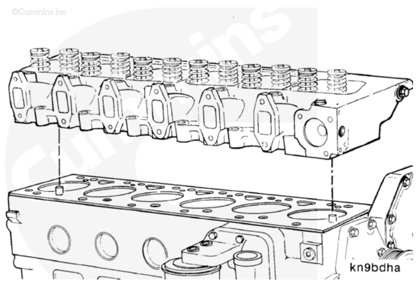
Remove the cylinder head gasket from the cylinder block.



B3.9, B5.9, and B4.5 Engines

Remove the cylinder head capscrews.





⚠ CAUTION ⚠

If removing the cylinder head with the injectors installed, be careful not to damage the tips of the injector. Do not set the cylinder head down on the combustion face with the injectors installed. Damage to the injector tips will result.

⚠ WARNING ⚠

The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

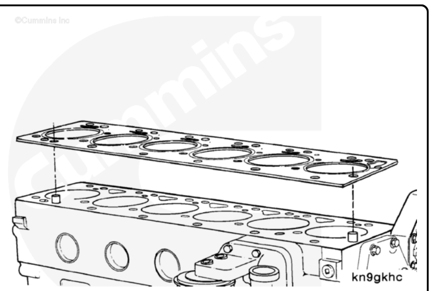
⚠ CAUTION ⚠

Do not lay the cylinder head on the combustion deck. This can cause damage to the cylinder head deck.

Remove the cylinder head from the cylinder block. Be sure the head is removed in a direct upward direction.

Cylinder Head Weight		
Cylinder Number	**Kg**	**lb**
4	36	79
6	51.3	113

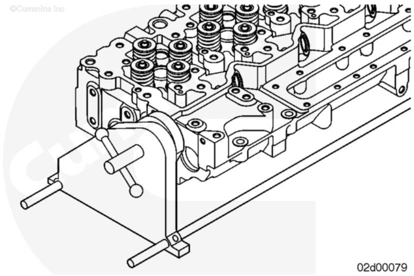
Remove the cylinder head gasket.



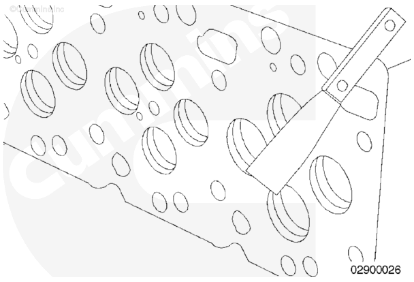
Initial Check

B4.5 RGT Engines

Install the cylinder head in the cylinder head holding fixture, Tool Number ST-583.

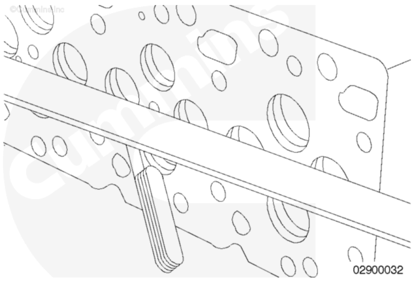


Scrape the gasket material and clean the combustion deck surfaces on the cylinder block and cylinder head.



Use a straightedge and a feeler gauge to inspect the cylinder head combustion surface for flatness.

Cylinder Head Flatness				
	mm		in	
End-to-End		0.305	MAX	0.012
Side-to-Side		0.076	MAX	0.003

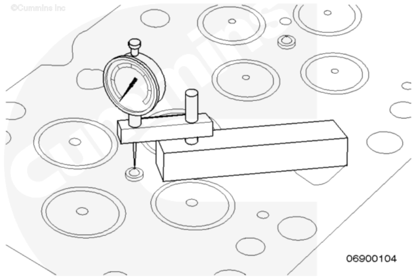


If out of specification, determine if the cylinder head can be resurfaced or if the cylinder head must be replaced by:

1. Measuring and recording valve depth
2. Measuring and recording injector protrusion.

If valve depth and injector protrusion specifications can be maintained, the cylinder head can be resurfaced. If the specifications can **not** be maintained, the cylinder head **must** be replaced.

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.



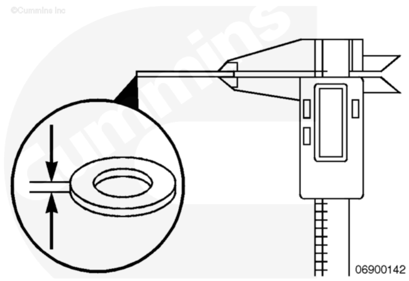
Rotate the depth gauge so that it is measuring the injector protrusion at the highest point on the injector.

Record the injector protrusion for each injector.

Injector Protrusion		
mm		in
2.45	MIN	0.096
3.15	MAX	0.124

Note : Do not use thicker or double stacked injector sealing washers to correct injector protrusion. This will cause misalignment of the high-pressure fuel connector.

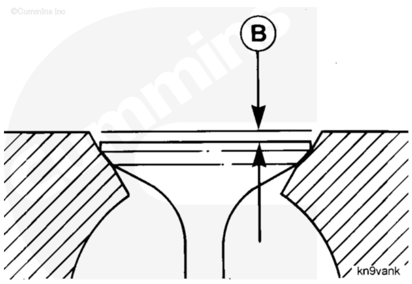
If the injector protrusion is out of specification, check the thickness of the injector sealing washer. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)



If the sealing washer is the correct thickness, check to make sure the injector bore is clean and free of debris. Also make sure that sealing washers are **not** 'stacked' in the injector bore.

If the injector protrusion is within specification, remove the injector. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.



Rotate the depth gauge so that it is measuring the valve recession into the cylinder head (B).

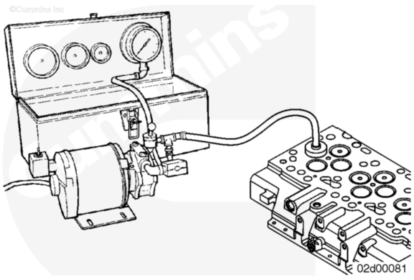
Record the valve depth for each valve.

Intake Valve Depth (Installed)		
mm		in
0.584	MIN	0.023
1.092	MAX	0.043

Exhaust Valve Depth (Installed)		
mm		in
0.965	MIN	0.038
1.473	MAX	0.058

Note : Valve depth can be increased slightly for resurfacing of the cylinder head by lapping the valves.

If a leaking valve is suspected, or if the cylinder head was recently rebuilt, vacuum test the valves and valve seats using valve vacuum tester, Tool Number 3824277, and cup, Part Number ST-1257-6. The vacuum **must not** drop more than 25.4 mm Hg [1.0 in Hg] in five (5) seconds.



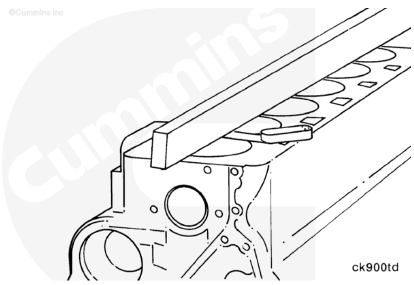
Note : If a vacuum tester is not available, with the valve removed, use a lead pencil or Dykem® marking pen to mark across the valve face. Install the valve in the valve guide. Hold the valve against the valve seat, and rotate the valve backward and forward three or four times. Correct contact against the valve seat will break the marks on the valve face.

Valve to Valve Seat Vacuum			
	kpa		in-hg
Used	51	NOM	15
New	69	NOM	20

If out of specification, disassemble the cylinder head and inspect for damaged valves and/or valve seats. Repair as necessary:

1. Clean the valve/valve seat and lap the valves.
2. Replace the damaged valve/valve seat, if available.
3. Replace the cylinder head.

Use a straightedge and feeler gauge to measure the overall flatness of the cylinder block.



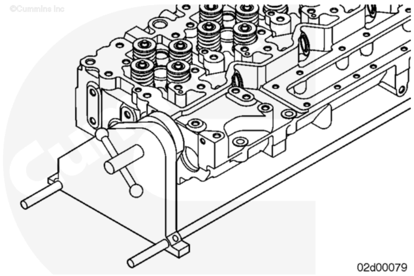
Cylinder Block Flatness				
	mm		in	
End-To-End		0.075	MAX	0.003
Side-To-Side		0.075	MAX	0.003

Inspect the combustion deck for any localized dips or imperfections.

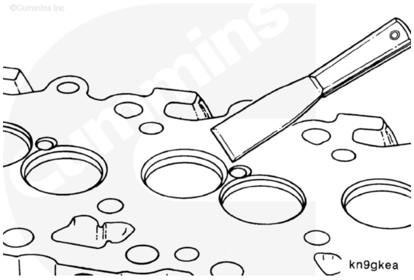
If out of specification, determine if the cylinder block can be resurfaced or if the cylinder block **must** be replaced. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-026.html)

B3.9, B5.9, and B4.5 Engines

Install the cylinder head in the cylinder head holding fixture, Part Number ST-583.



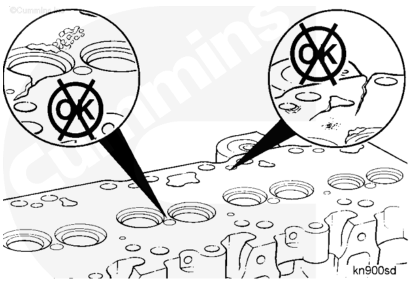
Scrape the gasket material and clean the combustion deck surfaces on the cylinder block and cylinder head.



Use a straightedge and a feeler gauge to inspect the cylinder head combustion surface for flatness.

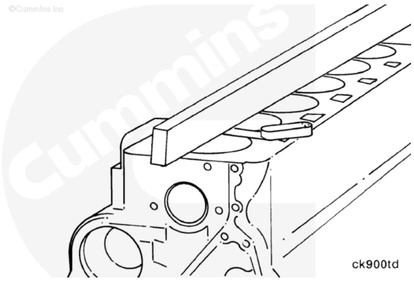
Cylinder Head Flatness (6 Cylinder)				
	mm		in	
End-to-End		0.305	MAX	0.012
Side-to-Side		0.076	MAX	0.003

Cylinder Head Flatness (4 Cylinder)				
	mm		in	
End-to-End		0.203	MAX	0.008
Side-to-Side		0.076	MAX	0.003



If the cylinder head combustion surface flatness is out of specification, the cylinder head can be resurfaced and an oversize head gasket used. Refer to Procedure 002-021 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-021-tr.html)

Use a straightedge and feeler gauge to measure the overall flatness of the cylinder block.

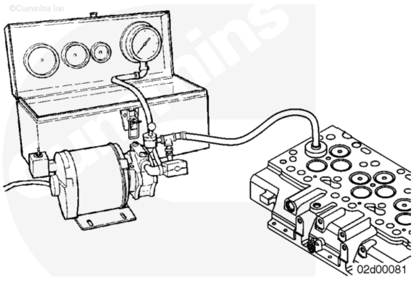


Cylinder Block Flatness				
	mm		in	
End-To-End		0.075	MAX	0.003
Side-To-Side		0.075	MAX	0.003

Inspect the combustion deck for any localized dips or imperfections.

If dips and/or imperfections are present, the cylinder head **must** be resurfaced. If resurfaced, an oversize head gasket **must** be used. Refer to Procedure 002-021 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-021-tr.html)

If a leaking valve is suspected, use valve vacuum test kit, Part Number 3824277, to test the valves and valve seats. The vacuum **must not** drop more than 25.4 mm Hg [1.0 in Hg] in five (5) seconds.



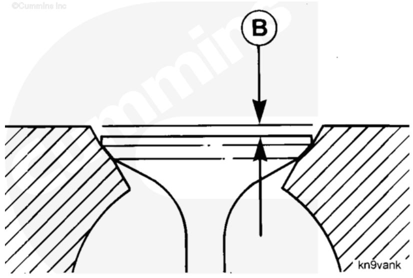
Note : If a vacuum tester is not available, with the valve removed, use a lead pencil or Dykem® marking pen to mark across the valve face. Install the valve in the valve guide. Hold the valve against the valve seat, and rotate the valve backward and forward three or four times. Correct contact against the valve seat will break the marks on the valve face.

Valve to Valve Seat Vacuum			
	mm-hg		in-hg
Used	457	MIN	18
New	635	MIN	25

If out of specification, disassemble the cylinder head and inspect for damaged valves and/or valve seats. Repair as necessary:

1. Clean and lap the valve/valve seat.
2. Replace the damaged valve.
3. Install valve seat inserts, if available.
4. Replace the cylinder head.

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.



Rotate the depth gauge so that it is measuring the valve recession into the cylinder head (B).

Measure the valve depth for each valve.

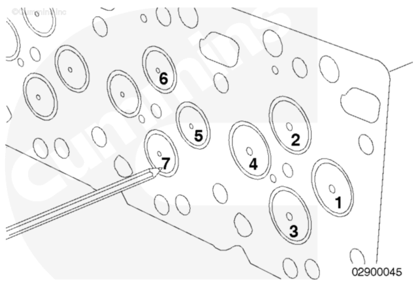
Valve Depth (Installed)		
mm		in
0.99	MIN	0.038
1.52	MAX	0.060

If the valve depth is **not** within specification during disassembly of the cylinder head, inspect the valve seats and valve for damage. If necessary, replace the valve and/or install valve seat inserts to achieve the correct valve depth. Information on valve seat insert installation is available in the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109.

Disassemble

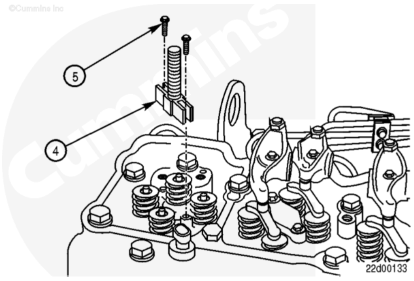
B4.5 RGT Engines

Mark the valves to identify their location.



Compress the valve springs using the valve spring compressor service tool, Part Number 3164329.

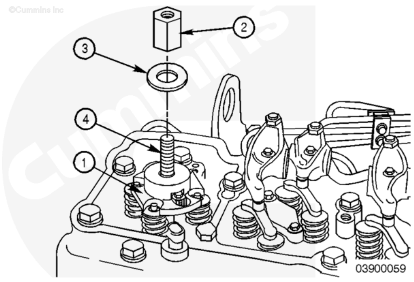
Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.



Tighten the capscrews (5).

Torque Value: 5 n•m [44 in-lb]

Note : The valves are not evenly spaced from the injector bore. It is important to align the slots in the valve spring compressor plate with the valve springs.



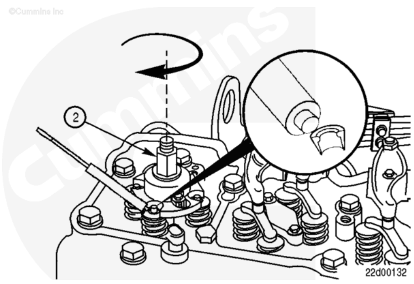
Apply anti-seize lubricant to the replacer screw (4) threads. Always read and follow label precautions.

Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

Install the washer (3) and nut (2) on the replacer screw (4).

⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.

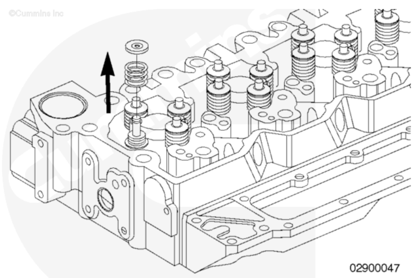


Turn the nut (2) **clockwise** to compress the valve springs.

Continue turning the nut (2) **clockwise** until the valve collets can be removed using a magnetic tool, such as the end of a magnetic screwdriver.

Remove the valve collets and the valve spring compressor service tool.

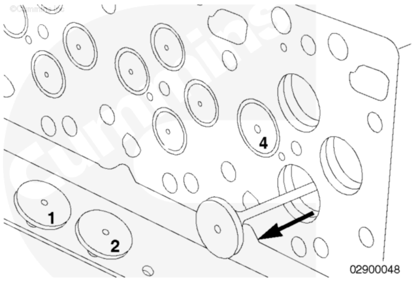
Remove the valve spring retainer and valve springs.



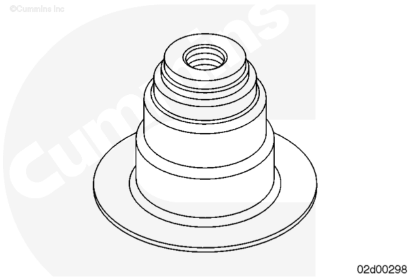
Note : Keep the valves in a labeled rack with the associated valve collets, spring retainers, and springs.

This will aid in assembling the components as a matched set.

Remove the valves.



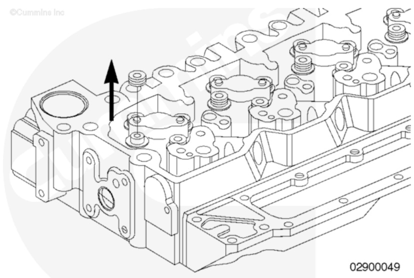
Note : Prior to removing the valve stems seals, note the color of the valve stem seal installed at each valve location. The same color valve stem seal must be installed when assembling the cylinder head.



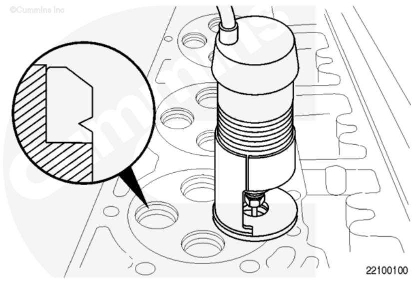
Colors Used:

- Green (used for exhaust valves)
- Yellow (used for intake and exhaust valves).

Use boot pliers, Part Number 3163293, to remove the valve stem seals.



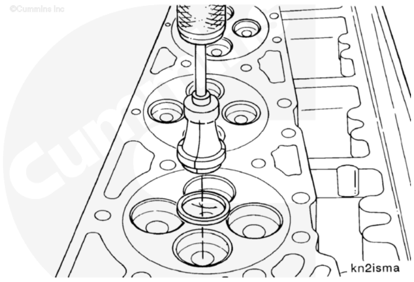
Note : Prior to removing the valve seat inserts, see the Initial Check and Clean and Inspect for Reuse sections in this procedure. The condition of the valve, the amount of recess, and the sealing of the valve on the seat insert all help determine whether or not a seat insert needs to be replaced.



1. If required, remove the valve seat inserts.
2. Inspect the valve insert-to-cylinder head contact area. A sufficient groove for the remover **must** exist.
3. If there is sufficient valve insert groove area, proceed to the next step.
4. If the valve insert groove area is **not** sufficient, use the valve seat insert cutting kit, Part Number 3376405, to create a sufficient groove.

Use the slide hammer remover, Part Number 3376617, with valve insert remover, Part Number 3165170, to remove the valve seats.

Note : Make certain that the valve insert remover assembly is perpendicular to the cylinder head when installed.



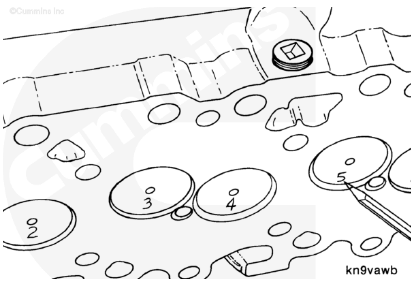
Insert the valve insert remover assembly into the valve insert and rotate the T-handle **clockwise** until the remover loosely grips the valve insert.

Position the valve insert remover assembly into the valve insert groove area. Tighten the T-handle firmly, allowing the remover to expand under the valve insert or into the cut groove.

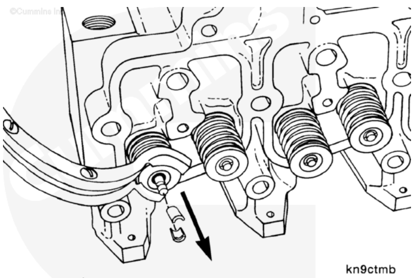
Strike the slide hammer remover against the top nut until the valve insert is removed. Turn the T-handle **counterclockwise** to release the valve insert from the remover.

B3.9, B5.9, and B4.5 Engines

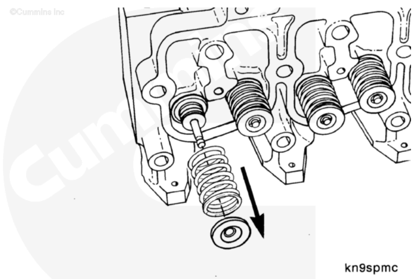
Mark the valves to identify their location.



Compress the valve spring using the valve spring compressor, Part Number 3375960, and remove the valve stem collets. Discard the collets.



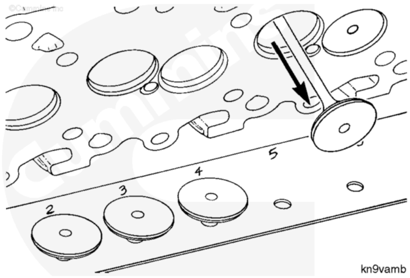
Release the valve spring and remove the spring retainer and spring.



Remove the remaining collets, retainers, springs, and valves.

Note : Keep the valves in a labeled rack for a correct match with companion seats while making

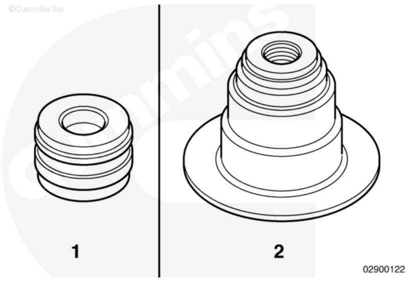
measurements.



Note : Prior to removing the valve stem seals, note the type of valve stem seal installed at each valve location. The same type of valve stem seal must be installed when assembling the cylinder head.

There are two types of valve stem seals used:

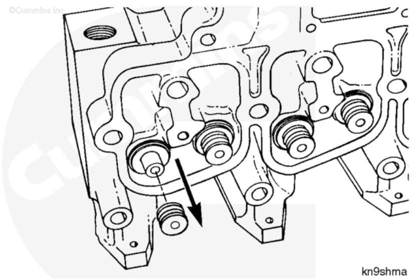
1. “Drive-On” Seal.
2. “Top-Hat” Seal.



Remove the valve stem seals.

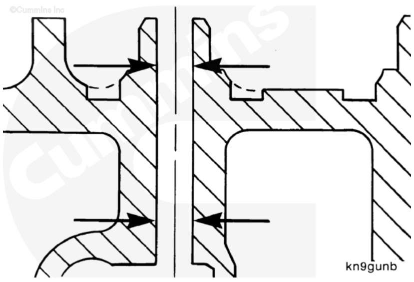
Use boot pliers, Part Number 3163293, to remove the valve stem seal.

Remove and discard the valve stem seals.



From production, the cylinder head does **not** have valve guide inserts or valve seat inserts. The valve guides and valve seats are machined integral parts of the cylinder head.

If servicing an engine in which the cylinder head was previously repaired with valve guide inserts or valve seat inserts, information is available in the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109.

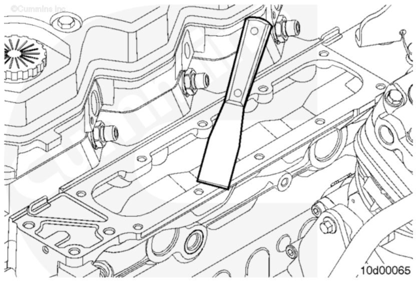


Clean and Inspect for Reuse

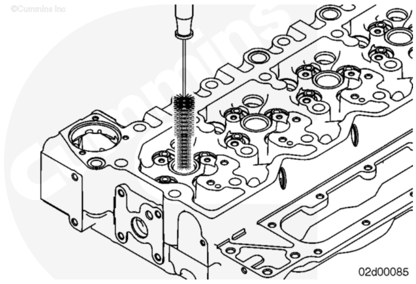
B4.5 RGT Engines

Note : Keep the gasket material and any other material out of the air intake.

If removed, clean the cylinder head sealing surfaces where the air intake manifold seals.

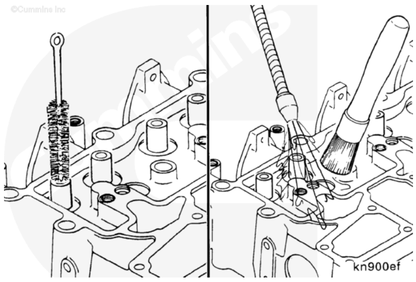


Use an injector bore brush, Part Number 3822510, or equivalent, to clean the carbon from the injector seat.



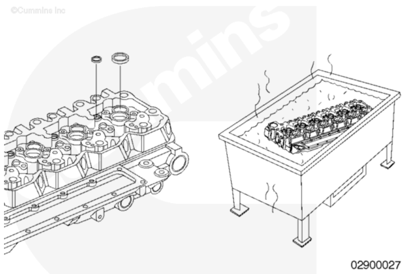
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use a bristle brush to clean the inside diameter of the valve guide bore and blow out with compressed air.

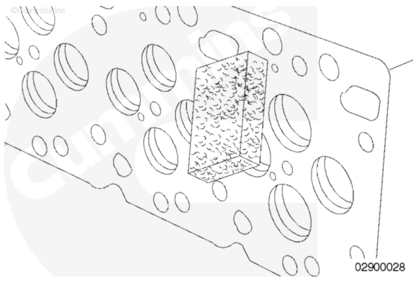
Note : Excessive deposits can be cleaned in an acid tank, but the expansion plugs **must** be removed first. Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/40/40-017-002.html)



If required, clean the buildup of deposits in the coolant passages.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



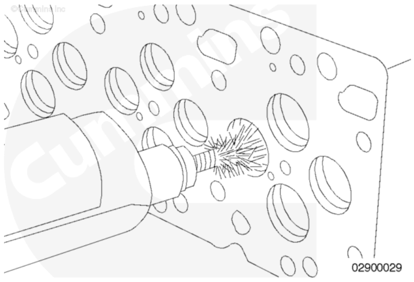
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Clean the cylinder head combustion deck with an abrasive hand pad, Part Number 3823258, or equivalent, and solvent.

⚠ WARNING ⚠

Wear protective eye covering while cleaning carbon deposits to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Contacting the valve seat with the wire wheel while it is spinning will damage the valve seat. If this occurs, new valve seats must be cut or new valve seat inserts must be installed.

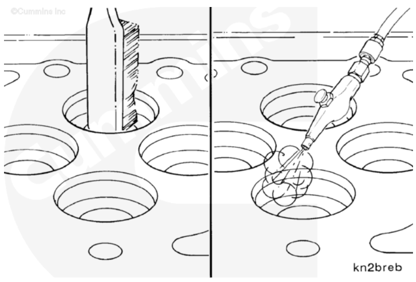
Inspect the area within 1/8-inch of the firing ring diameter. Any wear that can be felt with a fingernail within the 1/8-inch area is unacceptable, making the cylinder head **not** reusable. Wear beyond this 1/8-inch area will have no effect on future combustion sealing and the usability of the cylinder head.

Clean carbon deposits from the valve pockets with a high-quality steel wire wheel installed in a drill or die grinder.

Note : An inferior-quality wire wheel will lose steel bristles during operation, causing additional contamination.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

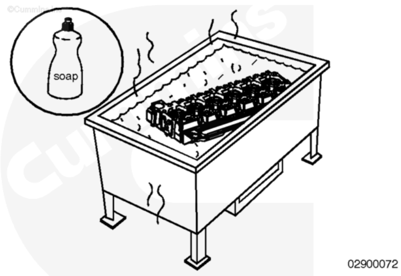
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush and solvent to clean the deposits from the valve seat insert bores, if it was necessary to remove the valve seat inserts.

Dry with compressed air.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

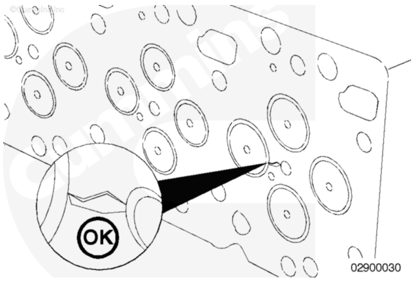


Wash the cylinder head in a hot, soapy water solution.

Rinse the cylinder head with clean water.

Dry the cylinder head with compressed air.

Cylinder Head Cracks - Reuse Guidelines



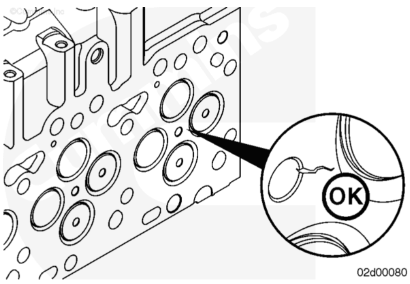
With the cylinder head cleaned, re-inspect the cylinder head for signs of cracks, fretting, and discoloration that would prohibit reuse.

To help identify cracks in the cylinder block, use the crack detection kit, Part Number 3375432.

Pay close attention to areas of the cylinder head that include:

- Injector bore
- Combustion face
- Valve seats
- Valve guides.

The reuse guidelines for a cylinder head with a crack extending from the injector bore to the intake valve seat are as follows:

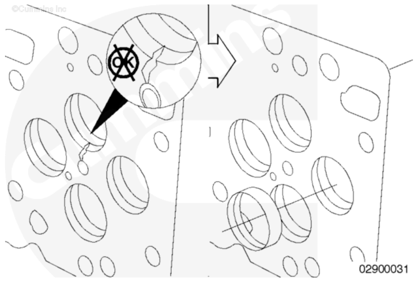


- If the crack does **not** extend into the valve seat, the cylinder head is reusable.

- If the crack extends into or through the valve seat, the cylinder head **must** be replaced.

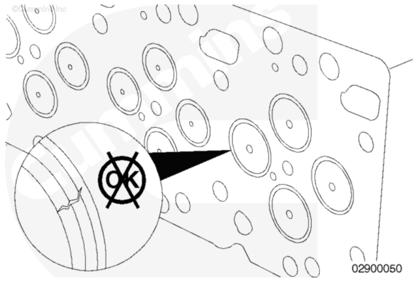
⚠ CAUTION ⚠

Failure to replace the cylinder head for a crack that extends into or through the valve seat bore will result in a valve seat insert falling out. Engine damage will result.



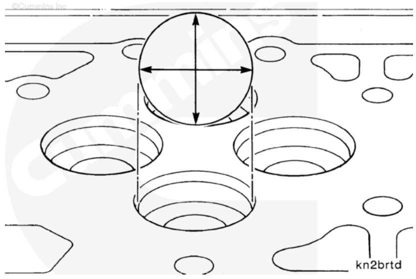
If still installed, inspect the valve seats for cracks or burn spots.

If the valve seat inserts are damaged, the cylinder head **must** be machined or replaced.



If the valve seat insert was removed in the Disassemble section, measure the inside diameter of the valve seat insert bore in the cylinder head.

Cylinder Head Insert Bore Inside Diameter (I.D.)		
mm		in
34.847	MIN	1.3719
34.863	MAX	1.3726



Note : Before cutting the cylinder head, verify valve seat inserts are available for the engine being serviced. If none are available, the cylinder head must be replaced.

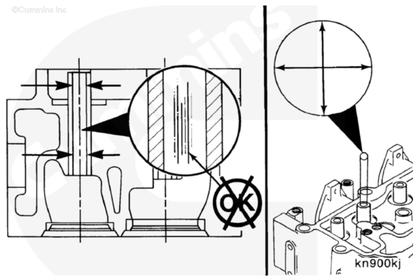
If out of specification, the valve seat insert bore can be oversized .254 mm (.010 in) and/or .508 mm (.020 in).

Use valve seat insert tool kit, Part Number ST257, with valve guide arbor, Part Number 3165184, to the cut cylinder head to accept oversize valve seat inserts. Use valve seat cutter, Part Number 3165183 (.254 mm (.010 in) or Part Number 3165184 (508 mm (.020 in).

Valve Guide - Reuse Guidelines

Inspect the valve guides for scuffing or scoring.

Measure the valve guide inner diameter (I.D.)

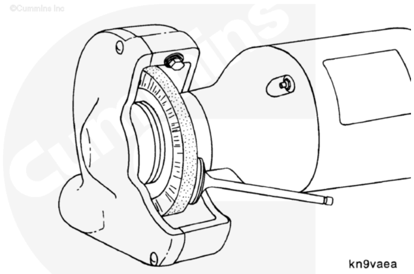


Valve Guide Bore Diameter		
mm		in
7.027	MIN	0.2767
7.077	MAX	0.2786

If the valve guide bore is worn larger than the maximum specified or if inspection reveals damaged valve guides, the cylinder head **must** be replaced.

⚠ WARNING ⚠

Wear protective eye covering when cleaning the valves with a wire wheel to reduce the possibility of personal injury.

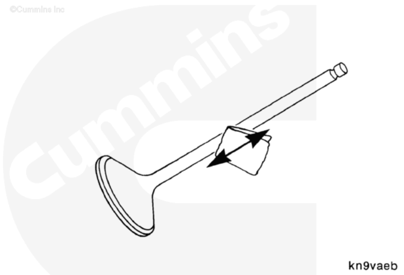


Clean the valve heads with a soft wire wheel.

Note : Keep the valves in a labeled rack to prevent mixing before taking measurements.

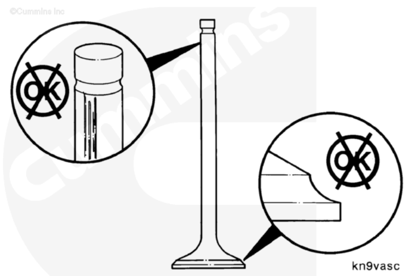
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Polish the valve stems with an abrasive pad, Part Number 3823258, and solvent.

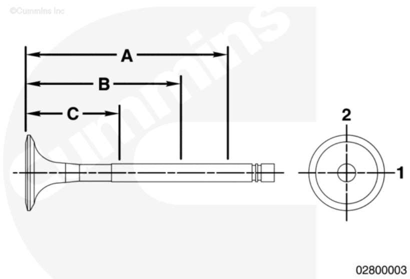
Valve - Reuse Guidelines



Inspect the valves for:

- excessive wear on the heads and stems
- excessive wear on the valve stem tips
- bends and distortion.

Inspect the valves for damage and the collet grooves for wear.



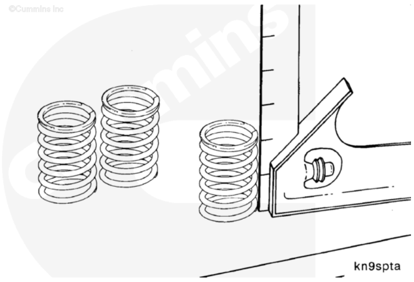
Measure the outside diameter of the valve stem.

Three measurements must be taken of each valve stem at 45 mm [1.77 in.], 70 mm [2.75 in.], 95 mm [3.74 in.] from the surface of the valve head.

Valve Stem Diameter		
mm		in
6.99	MIN	0.2752
7.01	MAX	0.2760

If the valves are damaged or the stems are worn smaller than the minimum specified, the valves **must** be replaced.

Inspect the valve springs.

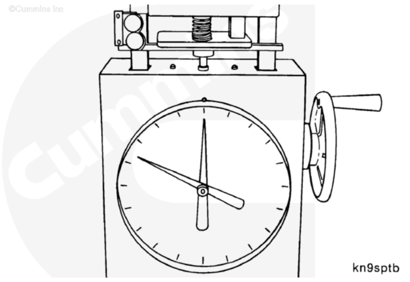


kn9spta

Use valve spring tester, Part Number 3375182, to compress the valve spring. A load of 320.8 to 358.8 N [72 to 80.7 lbf] is required to compress a spring to a height of 35.33 mm [1.39 in].

Note : If the valve spring is not within specification, a new valve spring must be used.

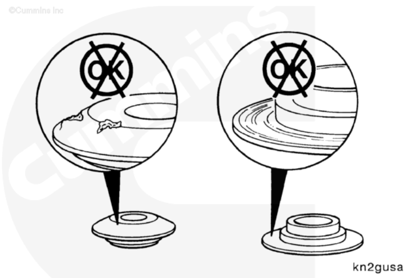
Note : Valve springs must be replaced in pairs under the same cross head. If one spring does not meet the specification, replace both valve springs under the same cross head.



kn9sptb

Inspect the valve spring retainers and valve collets for damage or worn areas.

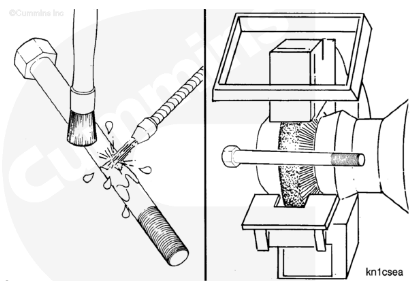
Discard and replace damaged and worn parts.



kn2gusa

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



kn1csea

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

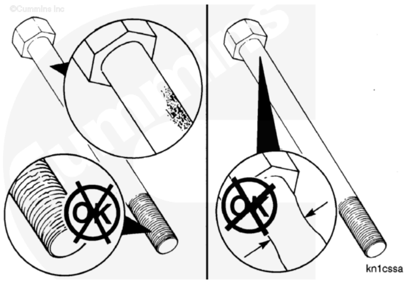
⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews. Component damage can occur.

Use a petroleum-based solvent to clean the capscrews.

Clean the capscrews thoroughly with a wire brush, soft wire wheel, or nonabrasive bead blast to remove deposits from the shank and threads.

Cylinder Head Capscrew - Reuse Guidelines



Inspect the cylinder head capscrews for damaged threads, corroded surfaces, or a reduced diameter (due to capscrew stretching).

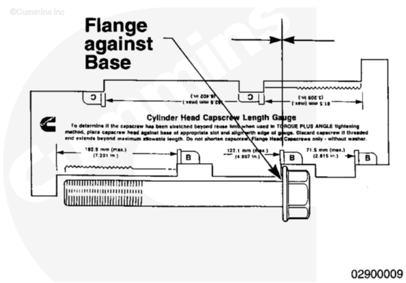
Do **not** reuse cylinder head capscrews under the following conditions:

- Visible corrosion or pitting exceeding 1 sq cm [0.155 sq in] in area.

Example:

- Acceptable is 9.525 x 9.525 mm [3/8 x 3/8 in]
- Unacceptable is 12.700 x 12.700 mm [½ x ½ in]
- Visible corrosion or pitting exceeds 0.12 mm [0.005 in] in depth
- Visible corrosion or pitting is located within 3.2 mm [1/8 in] of the fillet or thread
- Stretched beyond "free-length" maximum. See the measurement procedure below:

Free-Length Measurement



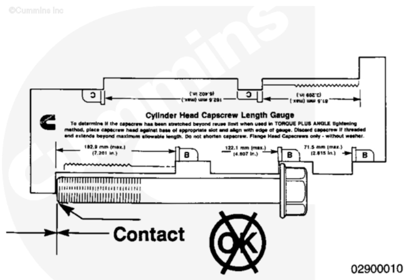
Note : If the capscrews are not damaged, they can be reused throughout the life of the engine, unless the specified free length is exceeded.

To check the capscrew free length, use capscrew length gauge, Part Number 316405. Place the head of the capscrew in the appropriate slot with the flange against the base of the slot.

Note : Most new cylinder head gaskets and upper engine gaskets include the capscrew length gauge, Part Number 3164057.

If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.

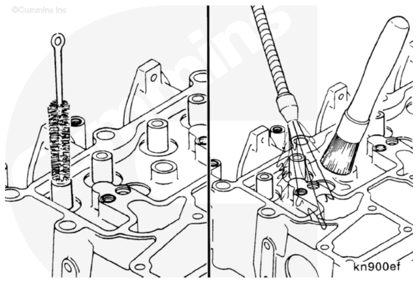
Cylinder Head Capscrew Free Length		
mm		in
152.1	MAX	5.99



B3.9, B5.9, and B4.5 Engines

⚠ WARNING ⚠

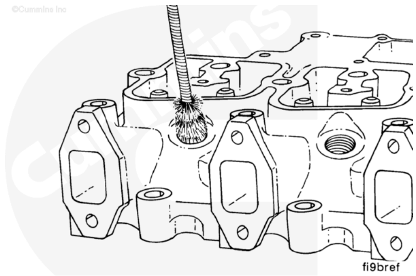
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the cylinder head with solvent.

Use a bristle brush to clean the inside diameter of the valve guide bore.

Use an injector bore brush, Part Number 3822509, or equivalent, to clean the carbon from the injector seat.



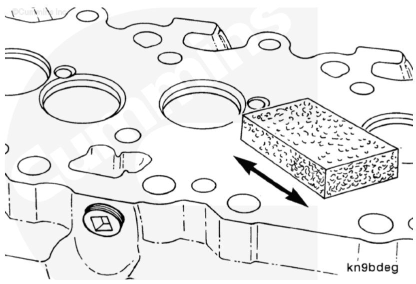
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use compressed air to dry the cylinder head.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

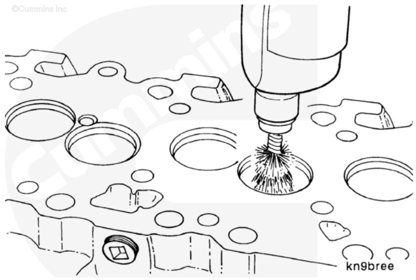
Clean the cylinder head combustion deck with an abrasive hand pad, Part Number 3823258, or equivalent, and solvent.

⚠ WARNING ⚠

Wear protective eye covering while cleaning carbon deposits to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Contacting the valve seat with the wire wheel while it is spinning will damage the valve seat. If this occurs, new valve seats must be cut or new valve seat inserts must be installed.



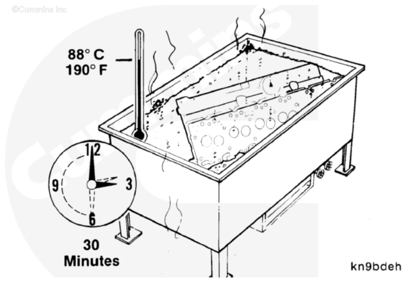
Inspect the area within 1/8-inch of the firing ring diameter. Any wear that can be felt with a fingernail within the 1/8-inch area is unacceptable, making the cylinder head **not** reusable. Wear beyond this 1/8-inch area will have no effect on future combustion sealing and the usability of the cylinder head.

Clean carbon deposits from the valve pockets with a high quality steel wire wheel installed in a drill or die grinder.

Note : An inferior-quality wire wheel will lose steel bristles during operation, causing additional contamination.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



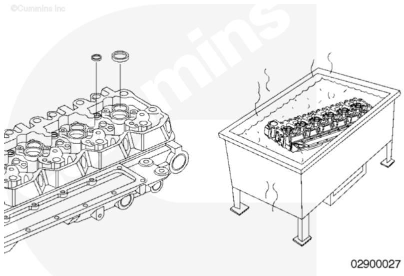
Wash the cylinder head in hot, soapy water solution.

After rinsing, use compressed air to dry the cylinder head.

Note : Excessive deposits can be cleaned in an acid tank, but the expansion plugs must be removed first.

Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/40/40-017-002.html)

Inspect the coolant passages for build up of deposits. If required, clean the coolant passages in an acid tank.



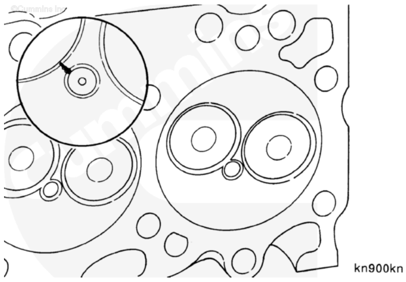
Cylinder Head Cracks - Reuse Guidelines

With the cylinder head cleaned, re-inspect the cylinder head for signs of cracks, fretting, and discoloration that would prohibit reuse.

To help identify cracks in the cylinder block, use the Crack Detection Kit, Tool Number 3375432.

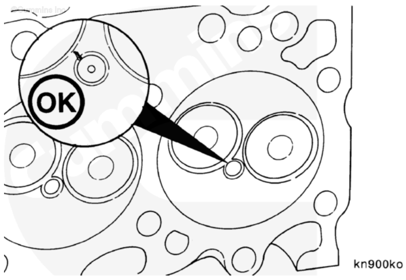
Pay close attention to areas of the cylinder head that include:

- Injector bore
- Combustion face
- Valve seats
- Valve guides.

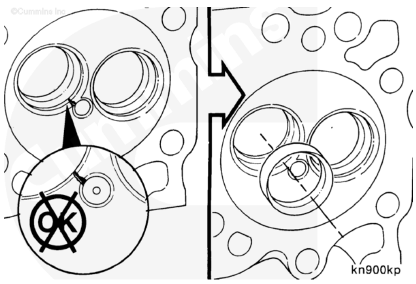


The reuse guidelines for a cylinder head with a crack extending from the injector bore to the intake valve seat are as follows:

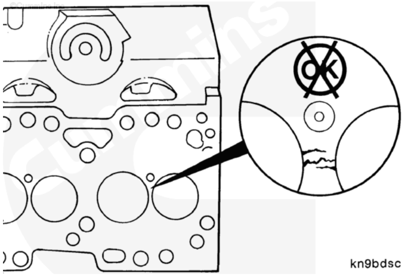
- If the crack does **not** extend into the valve seat, the cylinder head is reusable.



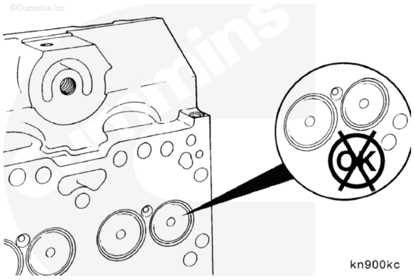
If the crack extends into or through the valve seat, the cylinder head **must** be repaired by installing a valve seat insert. Refer to the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109, for repair information.



Cracks between the valve seats are **not** acceptable and the cylinder head **must** be replaced.



Inspect the valves for indications of leakage or burning.

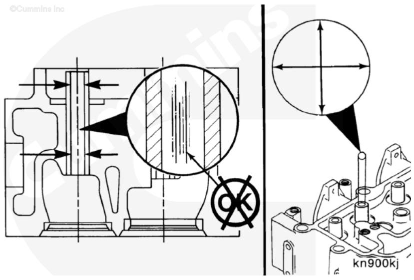


Depending on the severity, lapping of the valves/valve seats will be a sufficient repair.

If **not**, installation of valve seat inserts will be required.

Information on valve seat insert installation is available in the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109.

Valve Guide - Reuse Guidelines



Inspect the valve guides for scuffing or scoring.

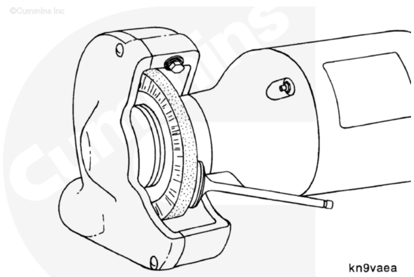
Measure the valve guide inner diameter (I.D.)

Valve Guide Bore Diameter		
mm		in
8.01	MIN	0.315
8.10	MAX	0.319

If the valve guide bore is worn larger than the maximum specified or if inspection reveals damaged valve guides, the cylinder head can be repaired with valve guide inserts. Information on valve guide insert installation is available in the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109.

⚠ WARNING ⚠

Wear protective eye covering when cleaning the valves with a wire wheel to reduce the possibility of personal injury.

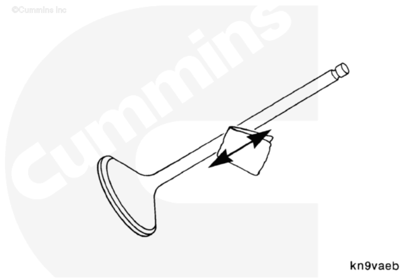


Clean the valve heads with a soft wire wheel.

Note : Keep the valves in a labeled rack to prevent mixing before taking measurements.

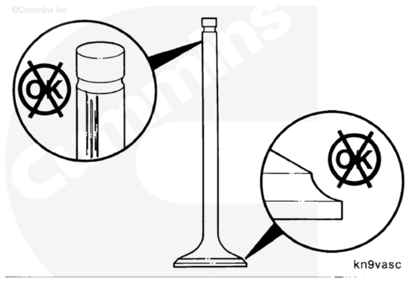
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Polish the valve stems with an abrasive pad, Part Number 3823258, and solvent.

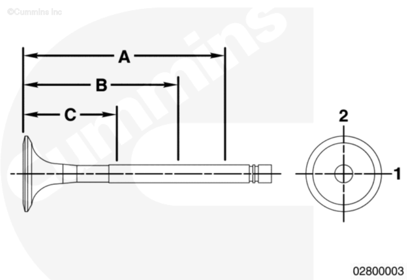
Valve - Reuse Guidelines



Inspect the valves for:

- excessive wear on the heads and stems
- excessive wear on the valve stem tips
- bends and distortion.

Inspect the valves for damage and the collet grooves for wear.



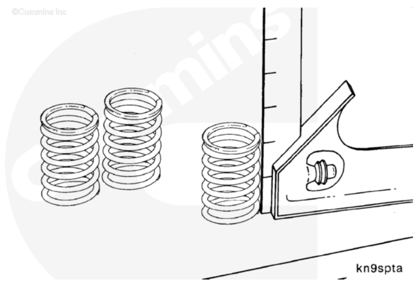
Measure the outside diameter of the valve stem.

Three measurements **must** be taken of each valve stem at 40 mm [1.57 in], 90 mm [3.54 in], and 140 mm [5.51 in] from the tip end.

Valve Stem Diameter		
mm		in
7.90	MIN	0.311
7.96	MAX	0.313

If the valves are damaged or the stems are worn smaller than the minimum specified, the valves **must** be replaced.

Valve Spring - Reuse Guidelines



Inspect the valve springs.

Measure the valve spring. Place a square adjacent to the spring and use a feeler gauge to measure the clearance at the top spring coil.

Measurements		
	mm	in
Approximate Free Length (L) 1991:	55.63	2.190

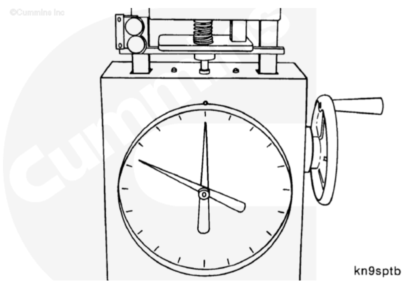
Measurements		
	mm	in
Maximum Inclination 1991:	1.00	0.039

Measurements		
	mm	in
Approximate Free Length (L) 1994:	60.00	2.362

Measurements		
	mm	in
Maximum Inclination 1994:	1.00	0.039

Use valve spring tester, Part Number 3375182, to compress the valve spring. A load of 289.13 to 321.16 N [65.0 to 72.2 lbf] (1991) and 359 to 397 N [80.7 to 89.2 lbf] (1994) is required to compress a spring to a height of 49.25 mm [1.94 in].

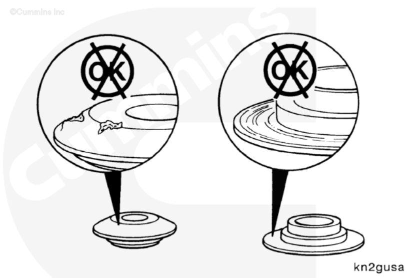
Note : If the valve spring is not within specification, a new valve spring must be used.



kn9sptb

Inspect the valve spring retainers and valve collets for damage or worn areas.

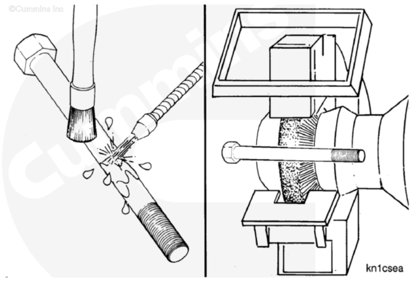
Discard and replace damaged and worn parts.



kn2gusa

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

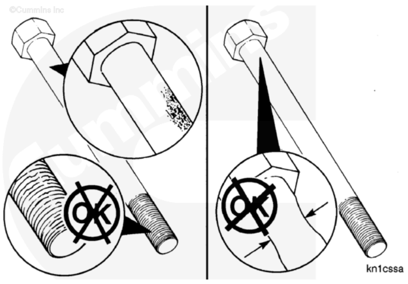
⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews. Component damage can occur.

Use a petroleum-based solvent to clean the capscrews.

Clean the capscrews thoroughly with a wire brush, soft wire wheel, or nonabrasive bead blast to remove deposits from the shank and threads.

Cylinder Head Capscrew - Reuse Guidelines



Inspect the cylinder head capscrews for damaged threads, corroded surfaces, or a reduced diameter (due to capscrew stretching).

Do **not** reuse cylinder head capscrews under the following conditions:

- Visible corrosion or pitting exceeding 1 sq cm [0.155 sq in] in area.

Example:

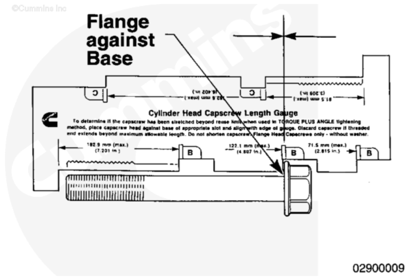
- Acceptable is 9.525 x 9.525 mm [3/8 x 3/8 in]
- Unacceptable is 12.700 x 12.700 mm [½ x ½ in]
- Visible corrosion or pitting exceeds 0.12 mm [0.005 in] in depth
- Visible corrosion or pitting is located within 3.2 mm [1/8 in] of the fillet or thread
- Stretched beyond "free-length" maximum. See the measurement procedure below:

Capscrew Length Gauge, Part Number 3823921

Free-Length Measurement

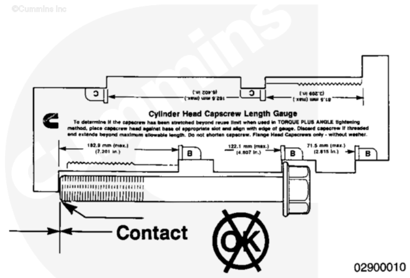
Note : If the capscrews are not damaged, they can be reused throughout the life of the engine, unless the specified “free length” is exceeded.

To check the capscrew free length, place the head of the capscrew in the appropriate slot with the flange against the base of the slot.

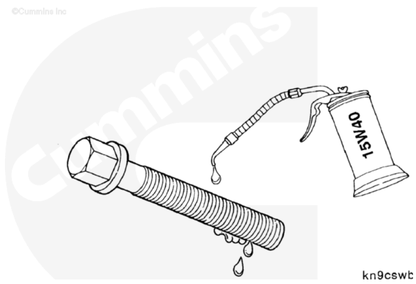


Capscrew Free Length				
	mm		in	
Short		71.5	MAX	2.815
Medium		122.1	MAX	4.807
Long		182.9	MAX	7.201

If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.



Immediately after cleaning and inspecting the capscrew, apply a film of clean engine oil to all capscrews that are to be reused.

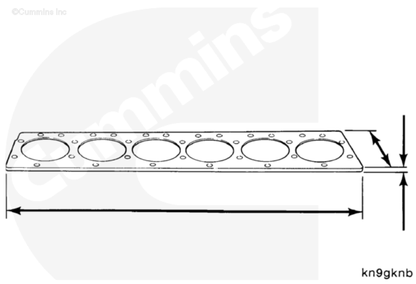


Pressure Test

B4.5 RGT Engines

If troubleshooting an internal coolant leak or coolant loss symptom, a cylinder head test fixture can be fabricated from a flat piece of steel or aluminum to pressure test the cylinder head.

See the following table for test fixture dimensions.



16 mm	Thickness	0.630 in

749 mm	Length	29.5 in
193 mm	Width	7.6 in

Note : Use the old cylinder head gasket as a pattern for drilling the capscrew holes.

Install the thermostat, water outlet connection and mounting capscrews.

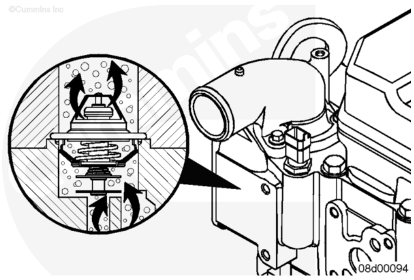
Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]

Install the engine coolant temperature sensor, located next to the water outlet connection.

Torque Value: 18 n•m [159 in-lb]

Note : The thermostat contains two check balls to vent air past the thermostat when closed. Install a rubber cap and hose clamp over the water outlet connection to prevent air from leaking through the check balls.



Install the cylinder head water test fixture.

1. Install a new head gasket.
2. Install the test plate.
3. Install the following:

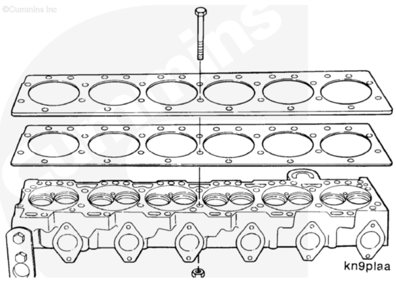
For four cylinder engines:

- 18 - 180-mm-long grade 12.9 flange head capscrews
- 18 - M12 x 1.75 hex flange nuts
- 36 - 12-mm washers.

For six cylinder engines:

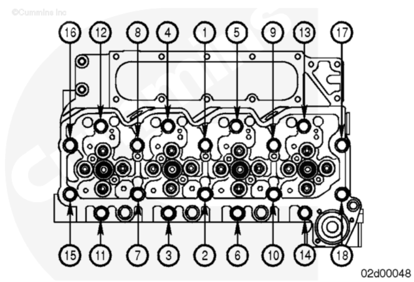
- 26 - 180-mm-long grade 12.9 flange head capscrews
- 26 - M12 x 1.75 hex flange nuts
- 52 - 12-mm washers.

Note : Place a washer between each capscrew and the head, and between each nut and test plate. This will prevent mutilation on the surface of the cylinder head.

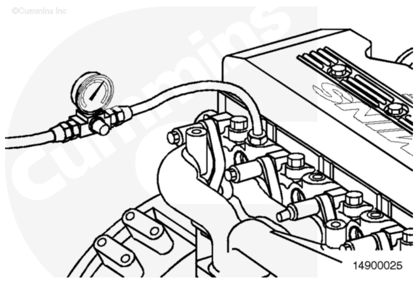


Use the illustrated sequence to tighten the four-cylinder nuts.

Torque Value: 80 n•m [59 ft-lb]

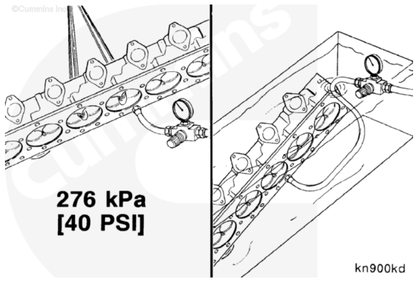


Service Tip: To apply air pressure to the cylinder head, remove one of the pipe plugs located on the exhaust side of the cylinder head. This is the same port used when the cylinder head is installed to check cylinder block coolant pressures.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly



Note : Make sure to plug or seal any open coolant ports before pressure testing the cylinder head.

Connect a regulated air supply hose, Part Number 3164231, to the cylinder head.

Apply air pressure.

Measurements		
	kpa	psi
Air Pressure:	276	40

Use a nylon lifting strap and a hoist to place the cylinder head in a tank of heated water.

Measurements		
	celsius	fahrenheit
Water Temperature:	60	140

Note : The cylinder head must be completely submerged in the water.

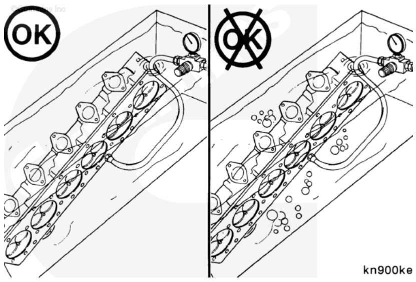
Inspect the head. Bubbles indicate an air leak.

If any bubbles exist, verify that the air leak is not coming from:

- any cup plugs or fittings installed in the cylinder head
- the test fixture or air line fittings

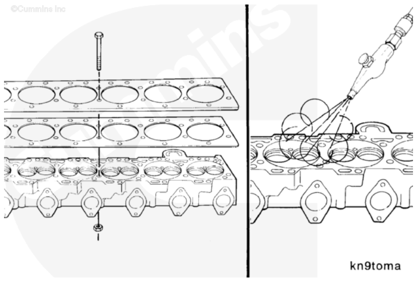
- water outlet connection.

If the above checks out OK and bubbles are present, the cylinder head leaks and it **must** be replaced.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



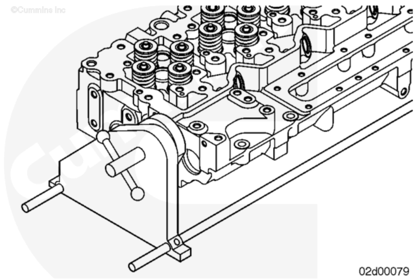
Remove the test fixture.

Use compressed air to dry the cylinder head.

Assemble

B4.5 RGT Engines

Install the cylinder head in the cylinder head holding fixture, Part Number ST-583.

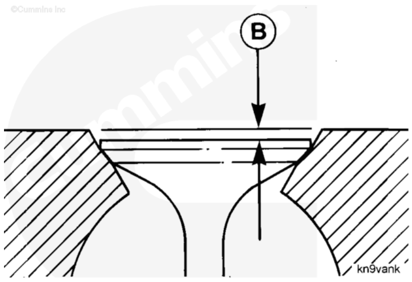


Note : When installing the valve seat inserts, the exhaust and intake valve seat inserts are not the same.

Valve seat angle:

- Intake 30-degrees
- Exhaust 45-degrees.

If new valve seat inserts are installed, check valve depth and perform a valve leak test. See the Initial Check of this procedure.

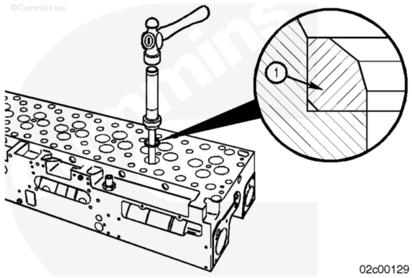


If the valve seat inserts were removed in the disassemble step, new inserts **must** be installed.

Note : The insert chamfer (1) must be installed toward the bottom of the counterbore.

Use valve seat installer, Part Number 3165171, to drive the intake and exhaust valve seat inserts into the counterbore.

Use a dead blow hammer with the seat drivers to install the new valve seat inserts.



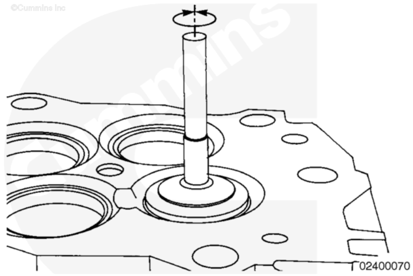
If new valve seat inserts were installed and/or the valve leakage was above specification, the valve seat/valve can be lapped.

Note : Lubricate the stems with 15W-40 engine oil before installing the valves.

Use a fine lapping compound, Part Number 3375805, or equivalent. Apply a thin and even coating on the valve.

Use a power or a hand suction lapping tool to provide pressure in the center of the valve.

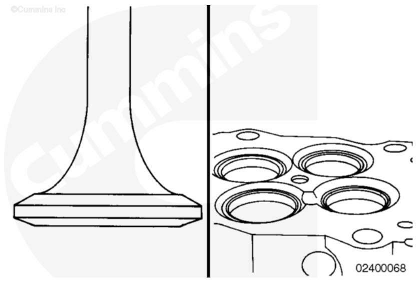
Turn the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, the valves, and the valve seats are not cleaned thoroughly.

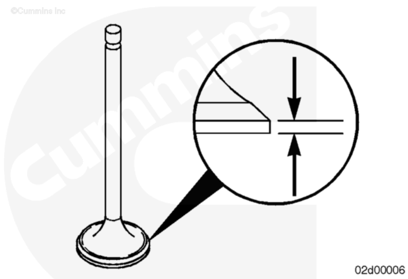
Clean the lapping compound from the parts.



If lapping of the valves was required, measure the rim thickness to determine if there is enough rim material left.

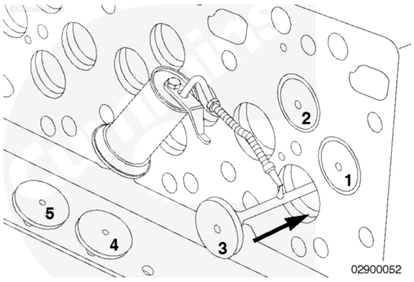
Valve Rim Thickness Limit		
mm		in
0.79	MIN	0.031

If the valve thickness is **not** within the limits, a new valve **must** be used.



⚠ CAUTION ⚠

Lubricate all the valve guide bores and valve stems with 15W-40 engine oil. Failure to lubricate the valve guides and valve stems can result in premature valve guide wear.



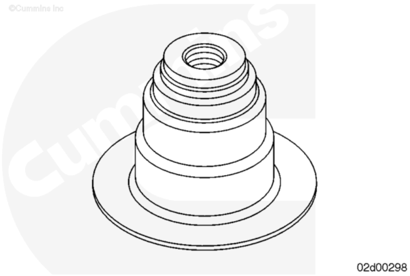
Lubricate the stems with 15W-40 engine oil before installing the valves.

Note : If installing the same valves as previously removed, make sure to install the valves in the same locations that the valves were removed.

Note : If the cylinder head will not be used right away, lubricate the valve stems with assembly lube, Part Number 3163087, or equivalent.

⚠ CAUTION ⚠

The same color valve stem seal must be installed in the same location as removed. Incorrect valve stem seals will result in excessive oil consumption and internal engine damage.



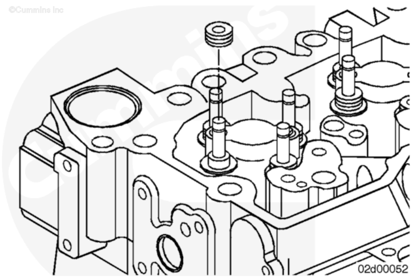
Install new valve stem seals of the same color as removed and in the same location.

There are two colors of valve stem seals used:

Colors Used:

- Green (Used for exhaust valves)
- Yellow (Used for intake and exhaust valves).

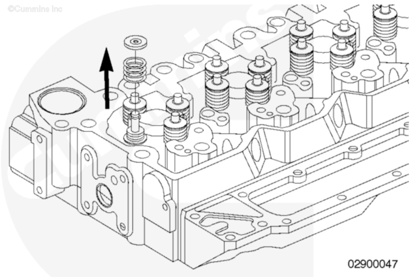
Install new valve stem seals using a valve stem installation tool, Part Number 3164055.



Note : The valve stem seals can be installed by hand. The installation tool will aid with installing the valve stem seals, but is not mandatory.

Use hand pressure to keep the valves from falling out during installation.

Install the valve spring retainer and valve springs.

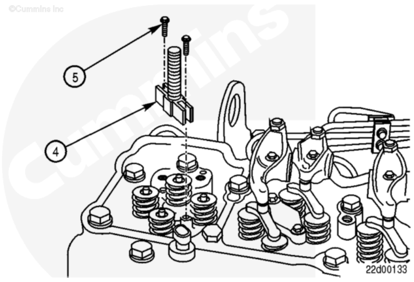


Compress the valve springs using the valve spring compressor service tool, Part Number 3164329.

Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.

Tighten the capscrews (5).

Torque Value: 5 n•m [44 in-lb]

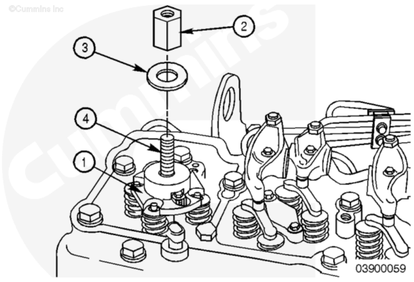


Note : The valves are not evenly spaced from the injector bore. It is important to align the slots in the valve spring compressor plate with the valve springs.

Apply anti-seize lubricant to the replacer screw (4) threads. **Always** read and follow label precautions.

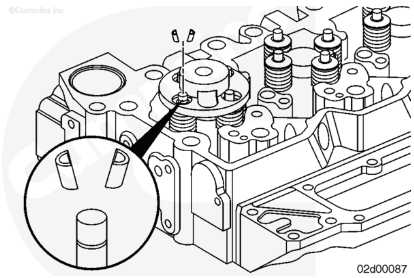
Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

Install the washer (3) and nut (2) on the replacer screw (4).



⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.



Compress the valve springs until the valve collets can be installed.

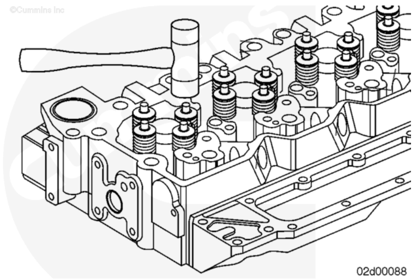
Install the valve collets.

Service Tip: Use assembly lube, Part Number 3163087, or equivalent, on the valve collets to help hold them in place until the valve spring compressor is released.

Remove the valve spring compressor service tool.

⚠ WARNING ⚠

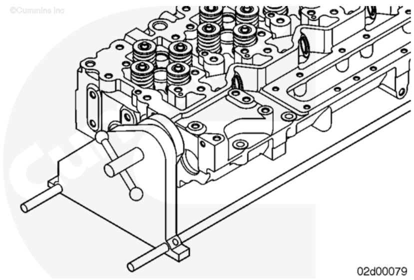
To reduce the possibility of personal injury, wear eye protection. If the collets are not correctly installed, they can fly out when the stems are hit with a hammer.



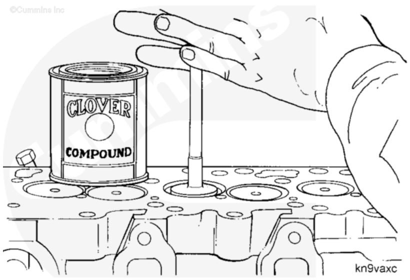
After assembly, hit the valve stems with a plastic hammer to make sure the collets are seated.

B3.9, B5.9, and B4.5 Engines

Install the cylinder head in the cylinder head holding fixture, Tool Number ST-583.



If during the inspection of the valve seats there were indications of leakage/burning and or the valve leakage during the leakage check was above specification, the valve seat/valve can be lapped.



Note : Lubricate the stems with 15W-40 engine oil before installing the valves.

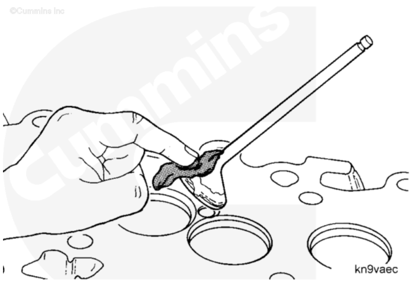
Use a fine lapping compound, Part No. 3375805, or equivalent. Apply a thin and even coating on the valve.

Use a power or a hand suction lapping tool to provide pressure in the center of the valve.

Turn the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat and the valve.

⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, the valves, and the valve seats are not cleaned thoroughly.

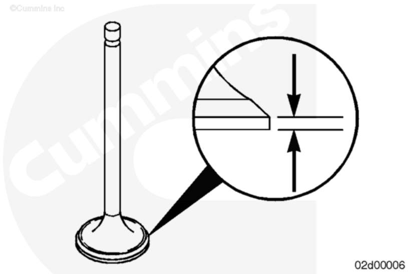


Clean the lapping compound from the parts.

Note : If after lapping the valves/seats there are still indications of leakage or burning, the installation of valve seat inserts will be required. Information on valve seat insert installation is available in the Alternative Repair Manual, B and C Series Engines, Bulletin 3666109.

If lapping of the valves was required, measure the rim thickness to determine if there is enough rim material left.

Valve Rim Thickness Limit		
mm		in
0.79	MIN	0.031

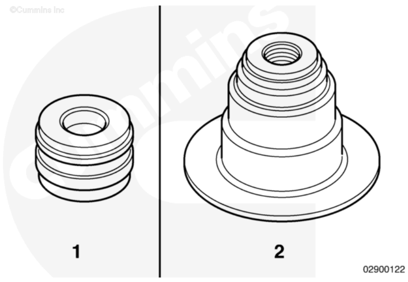


If the valve thickness is **not** within the limits, a new valve **must** be used.

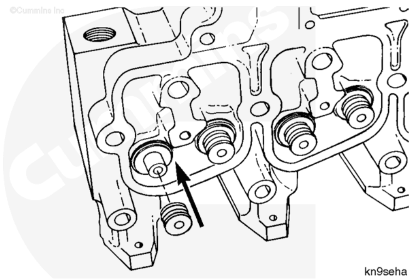
Note : Prior to installing the valve stem seals, make sure that the same type of valve stem seal is installed as was removed. The same type of valve stem seal must be installed when assembling the cylinder head.

There are two types of valve stem seals used:

1. “Drive-On” Seal.
2. “Top-Hat” Seal.

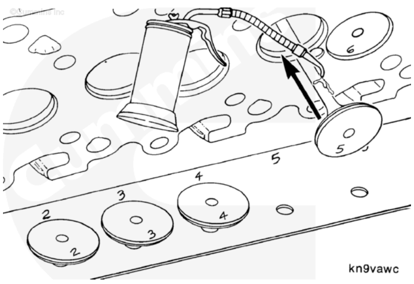


Install new valve stem seals. Press the valve stem seal onto the valve guide support by hand.



⚠ CAUTION ⚠

Lubricate all the valve guide bores and valve stems with 15W-40 engine oil. Failure to lubricate the valve guides and valve stems can result in premature valve guide wear.



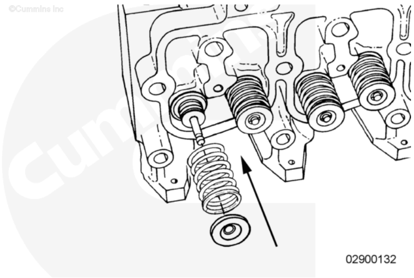
Lubricate the stems with 15W-40 engine oil before installing the valves.

Note : If installing the same valves as previously removed, make sure to install the valves in the same

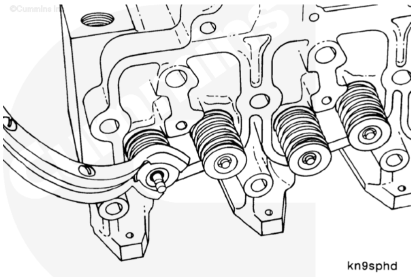
locations that the valves were removed.

Note : If the cylinder head will not be used right away, lubricate the valve stems with assembly lube, Part Number 3163087, or equivalent.

Install the valve spring retainer and valve spring.



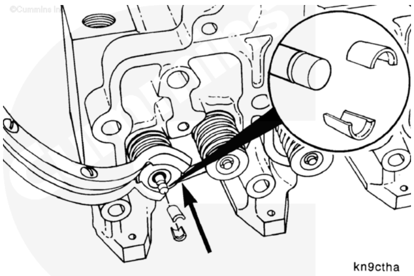
Use the valve spring compressor, Part Number 3375960, to compress the valve spring and retainer assembly.



Install new valve collets and release the spring tension.

Service Tip: Use assembly lube, Part Number 3163087, or equivalent, on the valve collets to help hold them in place until the valve spring compressor is released.

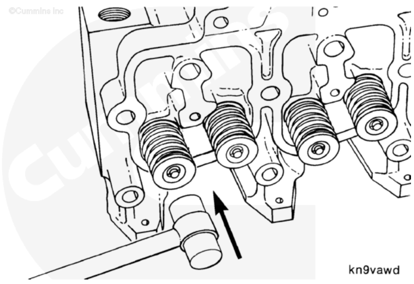
Remove the valve spring compressor service tool.



⚠ WARNING ⚠

Wear eye protection. If the collets are not correctly installed, they can fly out when the stems are hit with a hammer.

After assembly, hit the valve stems with a plastic hammer to make sure that the collets are seated.

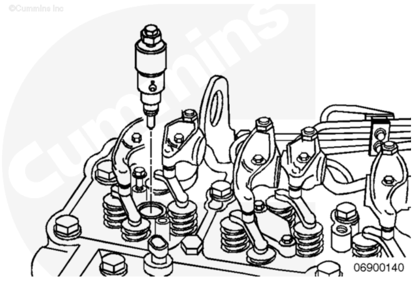


Measure

B4.5 RGT Engines

⚠ CAUTION ⚠

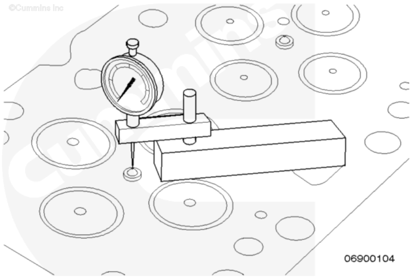
Improper injector protrusion can cause performance problems and high-pressure fuel leaks due to misalignment of the fuel connector.



Install the injectors with sealing washers into the cylinder head. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)

Measure the injector protrusion.

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.

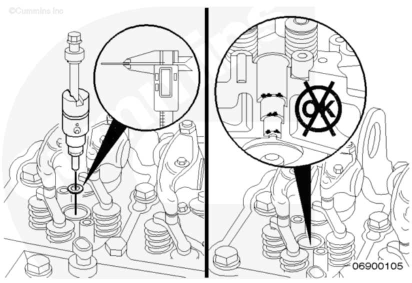


Rotate the depth gauge so that it is measuring the injector protrusion at the highest point on the injector.

Measure the injector protrusion for each injector.

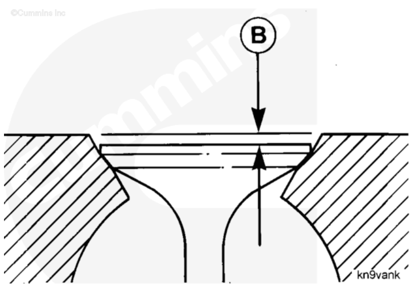
Injector Protrusion		
mm		in
2.45	MIN	0.096
3.15	MAX	0.124

If the injector protrusion is out of specification, check the thickness of the injector sealing washer. Refer to Procedure 006-026 (Injector) in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)



If the sealing washer is the correct thickness, check to make sure the injector bore is clean and free of debris. Also make sure that sealing washers are **not** 'stacked' in the injector bore.

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.



Rotate the depth gauge so that it is measuring the valve recession into the cylinder head (B).

Measure the valve depth for each valve.

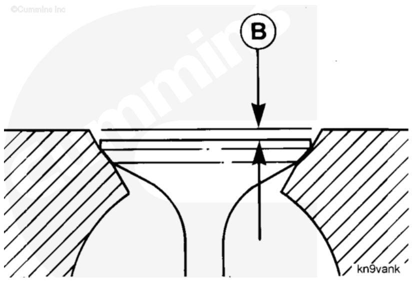
Intake Valve Depth (Installed)		
mm		in
0.584	MIN	0.023
1.092	MAX	0.043

Exhaust Valve Depth (Installed)		
mm		in
0.965	MIN	0.038
1.473	MAX	0.058

If the valve depth is **not** within specification, check if debris is preventing the valve from closing completely. If no debris is found, remove the valve and inspect the valve seat and valve face for damage.

B3.9, B5.9, and B4.5 Engines

Install depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.



Rotate the depth gauge so that it is measuring the valve recession into the cylinder head (B).

Measure the valve depth for each valve.

Valve Depth (Installed)		
mm		in
0.99	MIN	0.038
1.52	MAX	0.060

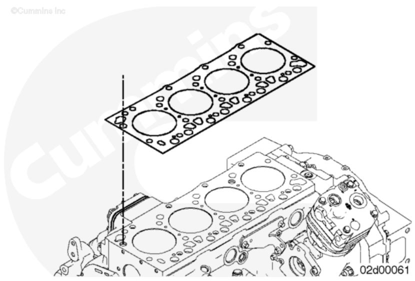
If the valve depth is **not** within specification, check if debris is preventing the valve from closing completely. If no debris is found, remove the valve and inspect the valve seat and valve face for damage.

Install

B4.5 RGT Engines

⚠ CAUTION ⚠

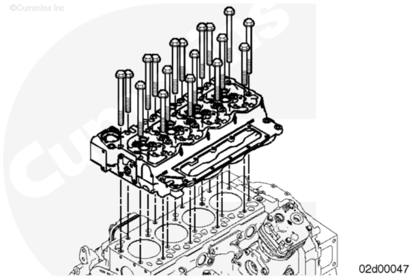
Make sure the gasket is correctly aligned with the holes in the cylinder block. Damage to the cylinder block can occur if the gasket is not aligned correctly.



Install the cylinder head gasket. Refer to Procedure 002-021 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-021-tr.html)

⚠ WARNING ⚠

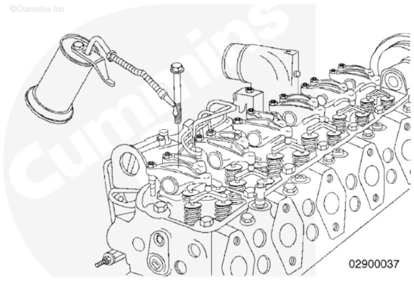
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Carefully put the cylinder head on the cylinder block and seat it onto the dowels.

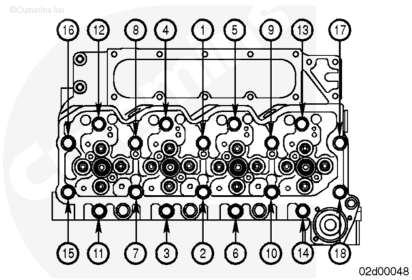
Lubricate the threads and under the heads on the cylinder head mounting capscrews with clean engine oil.

Install the capscrews and tighten finger-tight.



Use the illustrated sequence to tighten the cylinder head capscrews.

Tighten the capscrews.



Torque Value:

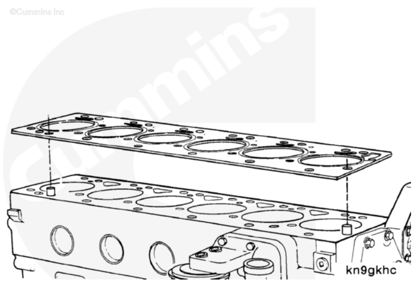
All Capscrews

1. 90 n•m [66 ft-lb]
2. 90 n•m [66 ft-lb]
3. Rotate 90-degrees clockwise.

B3.9, B5.9, and B4.5 Engines

⚠ CAUTION ⚠

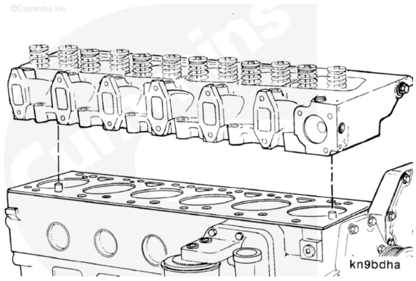
Be sure the gasket is correctly aligned with holes in the cylinder block. If the gasket is not aligned correctly engine damage can result.



Position a new cylinder head gasket over the dowels.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

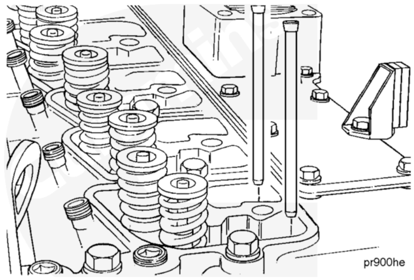


Carefully put the cylinder head straight down onto the cylinder block and seat it onto the dowels.

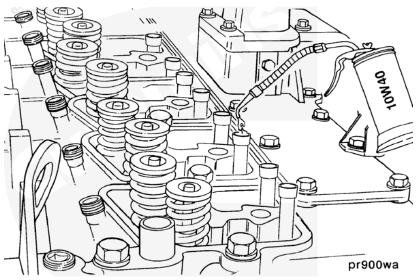
Cylinder Head Weight		
Cylinder Number	**Kg**	**lb**
4	36	79
6	51.3	113

Push Tubes - Installation

Position the push tubes into the valve tappets.

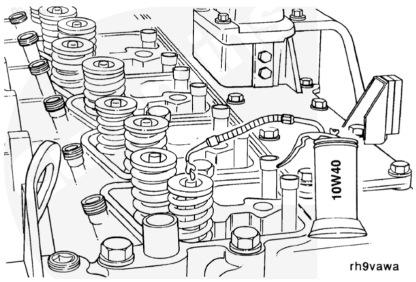


Lubricate the push tube sockets with clean lubricating engine oil.

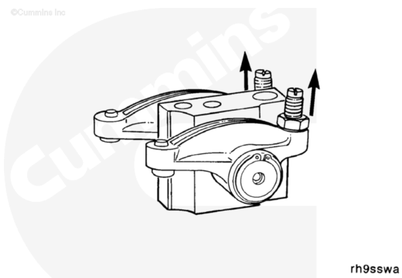


Rocker Levers - Installation

Lubricate the valve stems with clean lubricating engine oil.

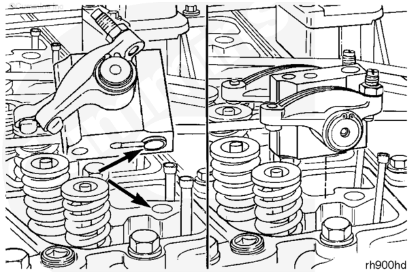


Completely loosen the rocker lever adjusting screws.



Note : The rocker lever pedestals are aligned with dowels.

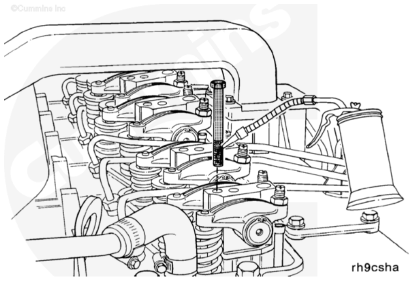
Install the pedestals.



Use capscrew length gauge, Part Number 3823921, to inspect all cylinder head capscrews for proper length.

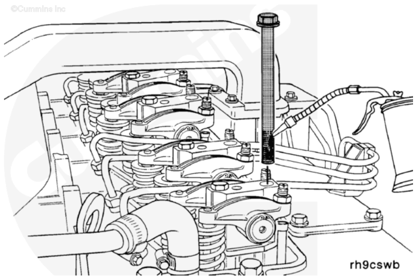
Lubricate the 8-mm pedestal capscrew threads and under the capscrew heads with clean lubricating engine oil.

Install the capscrews finger-tight.



Lubricate the 12-mm pedestal/head capscrew bolt threads and under the capscrew heads with clean lubricating engine oil.

Install the capscrews finger tight.

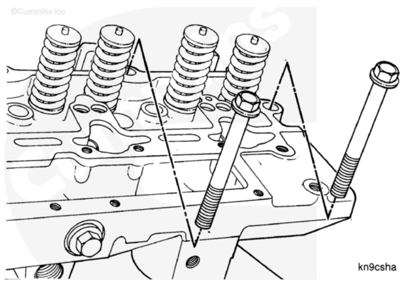


Lubricate the threads and under the heads on the remaining cylinder head capscrews with clean lubricating engine oil.

Install capscrews in the cylinder head and finger-tighten.

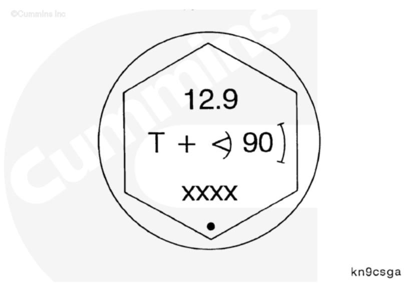
Note : Be sure to install the long capscrews into the holes underneath the injectors.

Note : Capscrews for the 1991 and later certification level engines are the same overall length, but have a longer threaded area.



⚠ CAUTION ⚠

Do not use pre-1991 certification level engine capscrews in a 1991 B Series or later certification level engine because pre-1991 capscrews do not have enough threads to provide enough thread engagement in the 1991 and later certification engines. Failure to do so can result in engine damage.



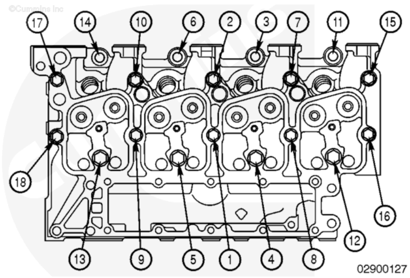
⚠ CAUTION ⚠

Capscrews for a 1991 B Series or later certification level engine can be used in a pre-1991 certification level engine because 1991 and later capscrews have enough threads to provide enough engagement in all certification level engines.

Note : The top of the cylinder head capscrew is identified with an angle marking. The cylinder head capscrews must be tightened by using the three-step "torque plus angle" method described as follows.

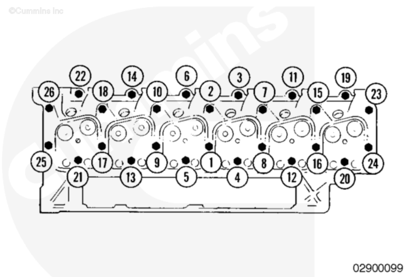
On a four cylinder engine, capscrew number 1 is located in between cylinders 2 and 3. The numbered sequence is the same as a six cylinder, but stops at capscrew number 18. Follow the numbered sequence for the four cylinder engine, and tighten all 18 capscrews.

Torque Value: 90 n•m [66 ft-lb]



Follow the numbered sequence for the six cylinder engine, and tighten all 26 capscrews.

Torque Value: 90 n•m [66 ft-lb]



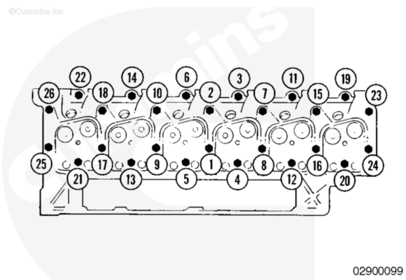
Four Cylinder

Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4,5,12, and 13).

Six Cylinder

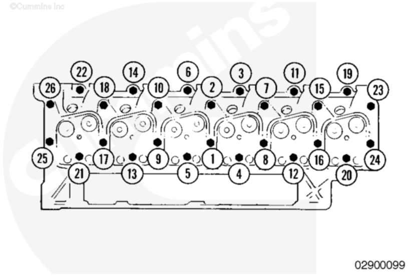
Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5,12, and 13, 20, and 21).

Torque Value: 120 n•m [89 ft-lb]

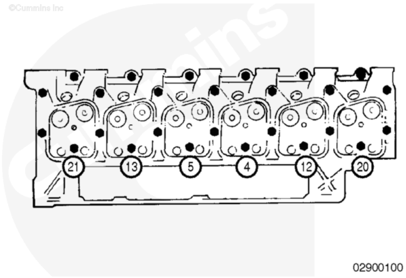


Tighten the short capscrews again (numbers 1, 2, 3, 6, 7, 8, 9, 10, 11, 14, 15, 16, 17, 18, 19, 22, 23, 24, 25, and 26) because of cylinder head relaxation and to obtain proper cylinder head torque requirements.

Torque Value: 90 n•m [66 ft-lb]



Tighten the long capscrews again because of cylinder head relaxation and to obtain proper cylinder head torque requirements.



Four Cylinder

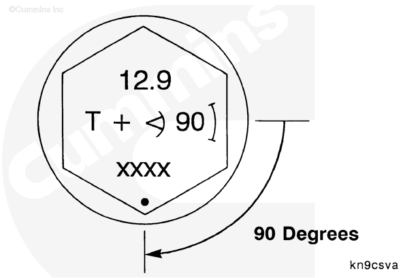
Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5, 12, and 13).

Six Cylinder

Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5, 12, and 13, 20, and 21).

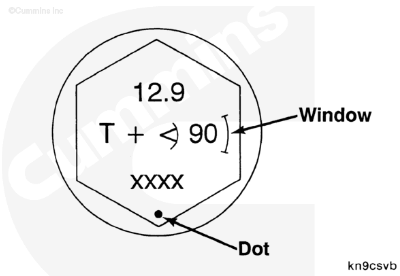
Torque Value: 120 n•m [89 ft-lb]

Follow the numbered sequence, and turn the capscrew 90 degrees as indicated on the capscrew head.

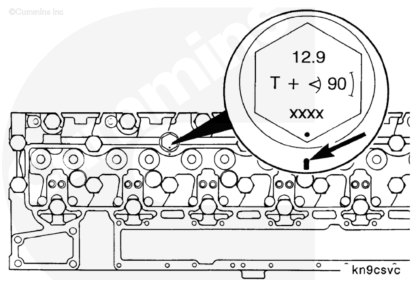


Torque Angle Gauge - 3/4-Inch Drive, Tool Number 3823878

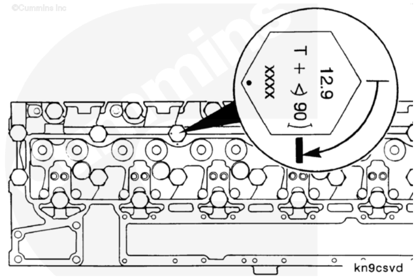
To turn the capscrew to the desired angle accurately, align the capscrew with the small “dot” and “window” that are marked on the capscrew head or use torque angle gauge - 3/4-inch drive tool, Part Number 3823878.



Mark the cylinder head adjacent to the dot on the capscrew head. This mark will serve as an indexing aid.

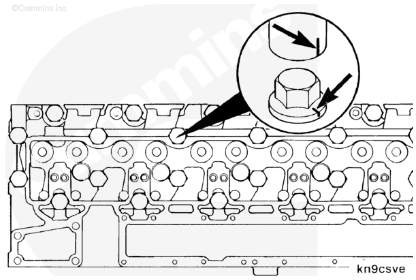


Rotate the capscrew until the mark that has been made on the cylinder head falls into the window on the capscrew head.

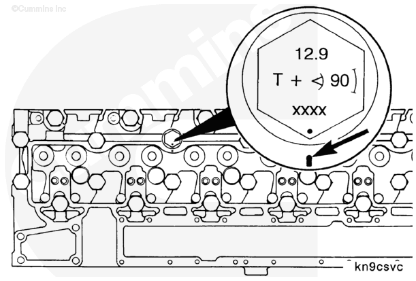


Service Tip:

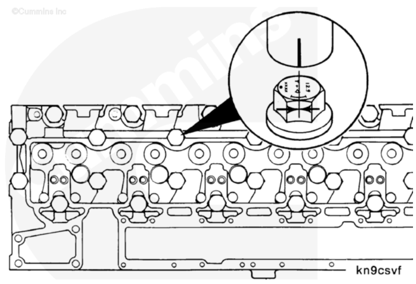
Use a permanent marker to mark the socket corresponding to one of the flats of the socket hex.



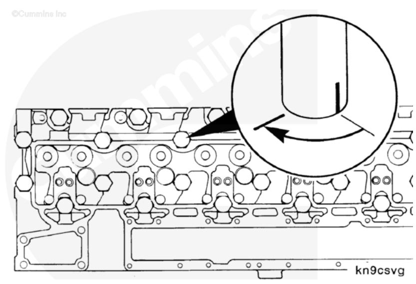
After the torque has been applied, mark the cylinder head at the location of the dot.



Position the socket on the capscrew such that the mark on the socket is at the same point as the window on the capscrew.

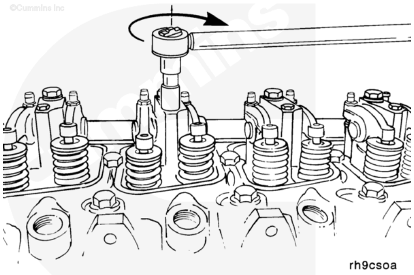


Turn the socket until the mark on the socket aligns with the mark on the cylinder head.



Tighten the 8-mm pedestal capscrews.

Torque Value: 24 n•m [212 in-lb]



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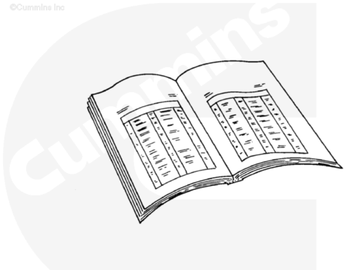
Finishing Steps



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50 C [120 F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

For B4.5 RGT engines, if the cylinder head was installed with the injectors installed, it may be necessary to loosen the injector to correctly install the fuel connectors. Failure to install the fuel connectors will result in excessive injector fuel leakage and poor engine performance.

- If **not** already installed, install the injectors. Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html)

- For B4.5 RGT engines, install the fuel connectors. Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-052-tr.html)
- Install the injector supply lines (high-pressure). Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Install the fuel supply lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html)
- For B4.5 RGT engines, install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- For B4.5 RGT engines, install the rocker lever assemblies and crossheads. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html) Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/40/40-002-001-tr.html)
- Adjust the overhead. Refer to Procedure 003-004 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html)
- If equipped, install the rocker lever housing. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-013-tr.html)
- Install the rocker lever cover gasket and rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/40/40-011-007-tr.html)
- If equipped, install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html)

Note : Omit the following steps if the engine is not equipped with the component or if it was not necessary to remove the component to remove the cylinder head.

- Install the thermostat housing and thermostat from the engine. Refer to Procedure 008-013 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-013-tr.html)
- Install the alternator bracket to the thermostat housing. Refer to Procedure 013-003 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-003-tr.html)
- Pivot the alternator towards the engine. Tighten the alternator link, mounting bolt, and water inlet connection capscrews. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-001-tr.html)
- Install the fan hub assembly. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-036-tr.html)
- Install the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Install the air intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-023-tr.html)
- Install the fuel filter and fuel filter head. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html) Refer to Procedure 006-017 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-017-tr.html)

Note : The follow components are required to be installed in order to complete the installation of the cylinder head.

- Connect the air crossover connection. Refer to Procedure 010-019 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- Connect all water and heater hoses attached to the cylinder head. See the equipment manufacturer service information.
- Install any OEM accessories attached to the cylinder head. See the equipment manufacturer service information.

- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-009.html)
- Operate the engine and check for leaks.

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